

5-1 5-2 (3) (5)
 5-3 5-4 5-6 (1) (3) (5)
 5-8 5-12 5-14 5-17
 5-18 5-19 5-21

5-1
 $T_c = \frac{2\pi}{2000\pi} = 10^{-3}$

5-1 (3) (4)

$W_S > W_c + W_M$

即 $W_M \leq W_c$

所以, 当 $W_S \geq 2W_c$ 时成立

$T_c \geq 2T_S$

$T_S \leq 0.5 \times 10^{-3} s$

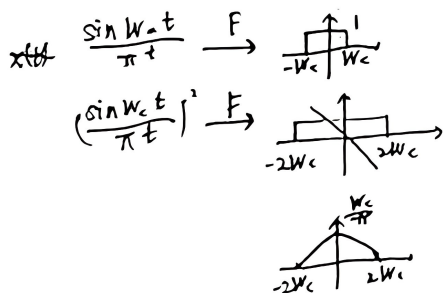
(1) (3) (4) 成立

5-2

(1) $W_M = 3000\pi$

$W_S = 6000\pi$

(3)



$W_S = 4W_c$

(5)

single

$W_M = 1000\pi + 2000\pi = 3000\pi$

$W_S = 6000\pi$

5-3

$W_M = 5\pi \quad W_S = \frac{2\pi}{5} = 10\pi$

(1) 会混叠

(2) $\frac{\pi}{T} = 5\pi = W_M$
 $X(f) = \sum_{k=0}^4 \frac{\sin k\pi f}{2k}$

5-5

$X_T(f) = \sum_{n=-\infty}^{\infty} X(nT) \text{Sa}[\pi(f - n/T)T] \frac{T}{\pi}$

$\therefore W_c = \frac{W_S}{2} = \frac{\pi}{T}$

$X_T(f) = \sum_{n=-\infty}^{\infty} X(nT) \text{Sa}[\pi(f - n/T)T]$

$X_T(kT) = \sum_{n=-\infty}^{\infty} X(nT) \text{Sa}[\pi(k - n)]$

仅 $k=n$ 时 $\text{Sa}[\pi(k-n)] \neq 0$

$X_T(kT) = X(kT)$

$\begin{cases} W_S = W_2 \\ A = T \\ W_b = W_2 = W_S = \frac{2\pi}{T} \\ W_2 + W_1 < W_a < W_1 \end{cases}$

5-12

$A = 10^{-3} s \quad W_S = 2k\pi$

$X_c(j\omega) = 10^{-3} X_d(j\omega) e^{j\omega t}$
 $W \leq k\pi$

5-6 (1)

$f_M = 400 \text{ Hz}$

$T \leq \frac{1}{800} s$

$T = \frac{1}{800} s$

(3)

$f_M = 300 \times 2 \text{ Hz}$

$= 600 \text{ Hz}$

$T = \frac{1}{1200} \text{ Hz}$

(5)

$f_M = 700 + \frac{300}{3}$

$= 200 \text{ Hz}$

$T = \frac{1}{400} s$

(1) $X_c(f)$ 实函数

(2) $X_c(f)$ 最大值为 $T = 10^{-3}$

(3) $\frac{3}{4}\pi \rightarrow 375 \text{ Hz}$

$\pi \rightarrow 500 \text{ Hz}$

$X_c(j\omega) = 0 \quad 750\pi \leq |\omega| \leq 1000\pi$

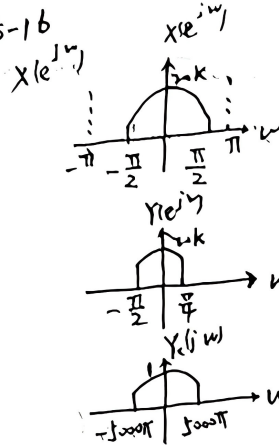
结合 $X_c(j\omega) = 0 \quad |\omega| \geq 1000\pi$

有 $X_c(j\omega) = 0 \quad |\omega| \geq 750\pi$

(4)

$X_c(j\omega) = X_c[j(\omega - 1000\pi)]$

5-16



5-8

$-W_1 + W_S \leq W_1$

$W_S \geq 2W_1$

$T_{max} = \frac{\pi}{W_1}$

$A = T_{max} = \frac{\pi}{W_1}$

$\begin{cases} -W_1 + W_S \leq W_a \leq W_1 \\ W_2 \leq W_b \leq W_1 + W_S \end{cases}$

$\Rightarrow W_a = W_1$
 $W_2 \leq W_b \leq 3W_1$

5-17 18 19 v1

5-17

$$H_c(j\omega) = \begin{cases} j\omega & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

$$\text{则 } H_d(e^{j\omega}) = \begin{cases} j\omega T & |\omega| \leq \omega_c T \\ 0 & |\omega| > \omega_c T \end{cases}$$

$$h_d[n] = T h_c(nT) = \frac{\omega_c T \cos(\pi n \omega_c T) - \sin(n \omega_c T)}{\pi n T}$$

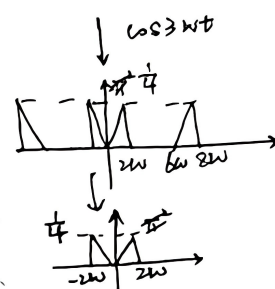
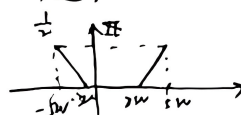
$$h_c(t) = \frac{\omega_c t \cos \omega_c t - \sin \omega_c t}{\pi t^2}$$

5-21

$$x(t) \cos 5\omega_c t$$



带通滤波后



5-18

$$H_d(e^{j\omega}) = \frac{1}{1 - \frac{1}{3} e^{-j\omega}}$$

$$H_c(j\omega) = H_d(e^{j\omega T}) = \frac{1}{1 - \frac{1}{3} e^{-j\omega T}} \quad (|\omega| < \frac{\pi}{T})$$

5-19.

$$g(t) \sin 400\pi t = x(t) \frac{\cos 200\pi t - \cos 600\pi t}{2}$$

$$= x_1(t) + x_2(t)$$



$$g(t) \sin 400\pi t = -\frac{1}{4} \sin 100\pi t + \frac{1}{2} \sin 200\pi t + \frac{1}{2} \sin 300\pi t$$

$$+ \frac{1}{2} \sin 400\pi t - \frac{1}{4} \sin 500\pi t - \frac{1}{2} \sin 600\pi t$$

$$y(t) = -\frac{1}{2} \sin 100\pi t + \sin 200\pi t + \sin 300\pi t$$

$$g(t) \sin 400\pi t = -\frac{1}{4} \sin 100\pi t + \frac{1}{4} \sin 300\pi t + \sin 400\pi t + \frac{1}{4} \sin 500\pi t - \frac{1}{4} \sin 700\pi t - \frac{1}{2} \sin 800\pi t$$

$$y(t) = -\frac{1}{2} \sin 100\pi t + \frac{1}{2} \sin 300\pi t$$